🧵 Python Strings: The Ultimate Homework

🎯 Objective

To master string manipulation using indexing, slicing, math operations, and loops—without using functions or `elif`.

❓ Problem 1: The Documentation (Docstrings)

1. **Task:** Create a multi-line string (docstring) that describes your favorite movie. It must span at least 3 lines.
2. **Action:** Print the string.
3. **Constraint:** Use triple quotes (`"""` or `'''`).

❓ Problem 2: String "Math" (Concatenation & Multiplication)

1. **Task:** Create two variables: `first_part = "Python"` and `second_part = "Rocks"`.
2. **Action:** * Create a third variable `full_sentence` by **concatenating** them with a space in between.
    - Create a fourth variable `excited_shout` by **multiplying** the `second_part` so it repeats 5 times.
3. **Action:** Print both `full_sentence` and `excited_shout`.

❓ Problem 3: The Secret Character (Indexing)

1. **Task:** Given the string `word = "DEVELOPER"`:
    - Use **positive indexing** to print the very first letter.
    - Use **negative indexing** to print the very last letter.
    - Use **negative indexing** to print the letter "O".

❓ Problem 4: The Knife (Slicing)

1. **Task:** Given the string `phrase = "Learning Python is fun"`:

- Slice and print only the word "Learning".
- Slice and print only the word "Python".
- Slice and print the word "fun" using **negative indices**.
- **Advanced:** Use a "step" in your slice to print every second character of the whole phrase.

The Snipping Tool (Slicing)

Task: Given the string `secret_msg = "00007James Bond00007"`:

1. **Slice** the string to print only "James Bond" (remove the zeros).
2. **Slice** the string to print only the first 5 characters.
3. **Slice** the string to print every **second** character (the step) of the entire string.
4. **Reverse** the string "James Bond" using only a slice trick (negative step).

❓ Problem 5: The Membership Test (`in` & `not in`)

1. **Task:** Create a variable `email = "student@school.com"`.
2. **Action:** * Use an `if/else` statement (no `elif`) and the `in` operator to check if the symbol `"@"` is in the email. Print "Valid email" or "Invalid email".
   - Use an `if/else` statement and the `not in` operator to check if the word `"spam"` is **not** in the email. Print "Safe message" if it's not there.

❓ Problem 6: The Character Speller (For Loops)

1. **Task:** Create a variable `hobby = "Coding"`.
2. **Action:** Write a `for` loop that iterates through the string and prints each character on a new line, but **only if** the character is not the letter "o".
3. **Constraint:** Use a single `if/else` inside your loop.

❓ Problem 7: Building a New String (Loop + Logic)

1. **Task:** Given the string `original = "Banana"`, create an empty string variable called `filtered`.
2. **Action:** Use a `for` loop to go through `original`.

    - If the character is "a", do nothing (or use `pass`).
    - Otherwise, "add" that character to your `filtered` string using concatenation.
3. **Result:** After the loop, print the `filtered` string (it should look like "Bnn").

🌟 Bonus Challenge: The Mirror

1. **Task:** Given a string `text = "Python"`, use **slicing with a negative step** to print the entire string backwards in one line of code.

8. The Vertical Speller (For Loops)

Task: Create a variable `word = "Python"`. **Action:** Write a `for` loop that iterates through `word`. Inside the loop, print each character followed by a hyphen. *Example Output:* `P- y- t-` ...

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9. The Vowel Filter (For Loop + If/Else)

Task: Given the string `sentence = "I am learning to code"`.

1. Create an empty string called `no_vowels`.
2. Write a `for` loop to go through the `sentence`.
3. Inside the loop, use an `if/else` statement:
    - **If** the character is "a", "e", "i", "o", or "u", do nothing.
    - **Else**, add that character to the `no_vowels` string. **Action:** Print `no_vowels` after the loop finishes.